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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSEZGYA00 - GENOME TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Answer Scheme

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer all questions
Each question carries 1 marks

1. **Continuous cell lines are often derived from:** **[U] / [CO1]**

Ans: b) Cancerous or immortalized cells

2. **Which concentration of Ethanol is most commonly used for surface sterilization in cell culture?** **[K] / [CO1]**

Ans: b) 70%

3. **The best stage of mitosis for preparing a karyotype is:** **[K] / [CO2]**

Ans: b) Metaphase

4. **Which reagent is commonly used to precipitate DNA from aqueous solution?** **[U] / [CO2]**

Ans: a) Isopropanol or ethanol

5. **What is the major advantage of CMA over Karyotyping?** **[An] / [CO2]**

Ans: b) It has a much higher resolution to detect microdeletions/duplications

6. **Satellite DNA is mainly** **[K] / [CO2]**

Ans: b) Highly repetitive DNA

7. In disease research, metabolomics can help identify: **[U] / [CO3]**

Ans: b) Altered metabolic pathways

8. Structural proteomics focuses on: **[U] / [CO3]**

Ans: b) Determining 3D structure of proteins

9. Which enzyme is used in RFLP analysis? **[U] / [CO2]**

Ans: b) Restriction enzyme

10. Which analysis in MEGA assesses the reliability of the tree branches? **[U] / [CO5]**

Ans: b) Bootstrap

[1x10 = 10]

Part B
Short Answer Questions
Answer any 14 questions
Each question carries 2 marks

11. Explain why a cell culture laboratory must maintain positive air pressure relative to the surrounding hallways. **[U] / [CO1]**

Ans: **Key point:** Positive air pressure ensures air flows *out* of the lab when doors open. ***(1 mark)***

Reason: This prevents outside airborne contaminants, microbes, and dust from entering the sterile laboratory environment. ***(1 mark)***

12. How is colony formation assessed in cell culture? **[U] / [CO1]**

Ans: By observing the growth pattern, morphology, and size of the colonies under a microscope. ***(1 mark)***

Health Indicator: Healthy adherent colonies have clear, defined edges; turbidity or fuzz indicates contamination. ***(1 mark)***

13. Explain the concept of passage number and its significance in cell culture maintenance. **[U] / [CO1]**

Ans: Definition: The total number of times a specific cell line has been subcultured/transferred. **(1 mark)**

Significance: High passage numbers can lead to genetic drift, phenotypic changes, or senescence, affecting experimental reproducibility. **(1 mark)**

14. What is the basic purpose of CRISPR technology in genetics? **[U] / [CO2]**

Ans: Core Purpose: It acts as a highly precise gene-editing tool (molecular scissors). ***(1 mark)***

Function: It allows scientists to target specific DNA sequences to knock them out or precisely rewrite the genetic code using a repair template. ***(1 mark)***

15. Describe why SNPs are considered useful markers in genomics. **[E] / [CO2]**

Ans: They are the most common type of genetic variation and are highly stable. ***(1 mark)***

Utility: Excellent for Genome-Wide Association Studies (GWAS) to link genetic variations with complex diseases and drug responses. ***(1 mark)***

16. Why is maintaining sterility especially important during tissue processing? **[U] / [CO1]**

Ans: Tissues removed from a biological specimen lose their immune system protection and are entirely vulnerable to microscopic contaminants. ***(1 mark)***

Since cell culture media is extremely rich in nutrients, any introduced bacteria or fungi will grow exponentially faster than the mammalian cells, outcompeting and killing the valuable primary tissue. ***(1 mark)***

17. Name the chemicals used for base-specific cleavage in Maxam–Gilbert sequencing **[K] /**

[CO3]

Ans: Dimethyl sulfate for Guanine (G), Hydrazine for Cytosine and Thymine (C and T). ***(1 mark)***
Hydrazine + salt for Cytosine (C only), and Piperidine for the actual cleavage. ***(1 mark)***

18. Define Quantitative Fluorescent PCR (QF-PCR). [K] / [CO2]

Ans: A rapid DNA amplification technique that uses fluorescently labeled primers to amplify Short Tandem Repeats (STRs). ***(1 mark)***
It is used for rapid prenatal diagnosis of chromosome abnormalities (e.g., Trisomy 21). ***(1 mark)***

19. State two advantages of NGS over traditional sequencing methods. [U] / [CO3]

Ans: Advantage 1: It is incredibly fast because it can read millions of DNA strands all at the same time, producing huge amounts of data very quickly. **(1 mark)**
Advantage 2: It is much cheaper when sequencing large amounts of DNA, and it is highly accurate at finding very rare genetic mutations. **(1 mark)**

20. Mention various applications of VNTRs in medical genetics. [E] / [CO2]

Ans: They are highly polymorphic and are used heavily in DNA fingerprinting for forensics and paternity testing. **(1 mark)**
Clinically, they are used for genetic linkage analysis to track how diseases are inherited within families, and to monitor bone marrow engraftment after transplants. **(1 mark)**

21. Describe how transcriptomics helps in understanding disease mechanisms. [E] / [CO3]

Ans: Transcriptomics measures active gene expression levels and identifies differentially expressed genes or alternative splicing events in a cell. **(1 mark)**

This data helps researchers classify cancers, understand host-pathogen interactions, and monitor how cells respond to specific drugs. **(1 mark)**

22. Why is WES focused on exons rather than introns? [E] / [CO3]

Ans: Whole Exome Sequencing (WES) specifically targets the protein-coding regions (exons) of the genome. **(1 mark)**
It focuses here because the vast majority (approximately 85%) of known disease-causing mutations occur within these functional coding regions, making it a highly efficient diagnostic tool. **(1 mark)**

23. Explain the difference between Global and Local sequence alignment. [U] / [CO5]

Ans: Global Alignment: Attempts to align the entire length of two sequences from end to end, which is best for sequences that are highly similar and of similar length. **(1 mark)**
Local Alignment: Searches for the most similar sub-regions or conserved motifs within two sequences, which is best for comparing sequences that are largely divergent. **(1 mark)**

24. How does the "Minimum ORF Length" parameter help reduce false positives? [U] / [CO4]

Ans: Random DNA sequences can frequently generate short, non-functional Open Reading Frames (ORFs) purely by chance. **(1 mark)**
Setting a minimum length filter eliminates these tiny, random reading frames, leaving only the sequences that have a statistically significant probability of coding for a real functional protein. **(1 mark)**

25. What specific information is found in the 'FEATURES' table of a GenBank record? [U] / [CO4]

Ans: The FEATURES table contains detailed biological annotations about the sequence. **(1 mark)**

It includes the exact coordinates of genes, coding sequences (CDS), regulatory regions, and the resulting protein translations. **(1 mark)**

26. Explain the purpose of the Variant Effect Predictor (VEP) in Ensembl. [U] / [CO4]

Ans: VEP is a bioinformatics tool used to analyze and predict the functional consequences of genetic variants (such as SNPs or indels). **(1 mark)**

It determines how a specific mutation will affect a transcript or protein (e.g., causing a frameshift or missense mutation) and cross-references clinical databases to assess its potential to cause disease. **(1 mark)**

[2x14 = 28]

Part C
Short Essay Questions
Answer any 8 questions
Each question carries 4 marks

27. Explain the major components of media (salts, glucose, amino acids, vitamins, serum, antibiotics) and their roles. [U] / [CO1]

Ans: Basal Medium (Salts, Glucose, Amino Acids, Vitamins): Provide essential nutrients, energy, and maintain osmotic pressure/pH for basic cell survival. ***(1.5 marks)***

Serum (e.g., FBS): Provides crucial growth factors, hormones, and attachment factors required for active cell proliferation. ***(1.5 marks)***

Antibiotics: Prevent bacterial and fungal contamination during culture establishment. ***(1 mark)***

28. Explain the step-by-step procedure for starting a primary cell culture from a biological specimen. [U] / [CO1]

Ans: Collection & Mincing: Harvesting tissue safely and mincing it into < 1 mm^3 pieces. ***(1 mark)***

Disaggregation: Using mechanical forces or enzymatic digestion (Trypsin/Collagenase) to break the extracellular matrix and release single cells. ***(1.5 marks)***

Filtering & Seeding: Filtering out large debris, washing the cells, and seeding them into a culture vessel with appropriate media. ***(1.5 marks)***

29. Explain the function of Taq polymerase in PCR and its importance in the amplification process.

Ans: Definition/Source: A thermostable DNA polymerase isolated from *Thermus aquaticus*. ***(1 mark)***

Importance in Denaturation: It can withstand the extreme heat (94–98°C) required to separate DNA strands without denaturing itself. ***(1.5 marks)*** **[U] / [CO2]**

Role in Extension: Synthesizes the new DNA strand at 72°C by adding dNTPs in the 5' to 3' direction. ***(1.5 marks)***

30. Analyse the principles and applications of quantitative PCR (qPCR), explaining how it enables precise measurement of initial DNA or RNA quantities. **[An] / [CO2]**

Ans: Principle: Uses fluorescent dyes (SYBR Green) or probes (TaqMan) to monitor DNA amplification continuously in real-time. **(2 marks)**

Application: Enables precise measurement and absolute/relative quantification of the initial amount of target DNA/RNA, unlike traditional end-point PCR. ***(2 marks)***

31. Describe the role of next-generation sequencing in diagnostic gene panels and evaluate its effectiveness. **[E] / [CO3]**

Ans: Role: Allows simultaneous sequencing of a predefined group of genes associated with specific diseases. ***(1 mark)***

Advantages: High throughput, fast, cost-effective per base, and excellent for hereditary disease screening. ***(1.5 marks)***

Limitations: Strictly limited to the pre-selected genes; cannot detect mutations in novel genes not included in the panel design. ***(1.5 marks)***

32. Evaluate whether Targeted Exome Sequencing is suitable for prenatal or pediatric genetic testing. **[E] / [CO3]**

Ans: Suitability: Highly suitable *only if* the target genes/phenotypes are previously known. ***(1 mark)***

Advantages: High coverage depth, reliability for low-frequency variants, and faster/cheaper than Whole Genome Sequencing. ***(1.5 marks)***

Limitations: Not suitable for diagnosing entirely new, complex, or unknown genetic disorders where the causative gene is outside the targeted exome. ***(1.5 marks)***

33. Evaluate the role of primer design in determining the success and specificity of Sanger sequencing. **[E] / [CO3]**

Ans: Primers dictate the exact starting point and specificity of the sequencing reaction. Poor primers lead to unreadable chromatograms. ***(1.5 marks)***

Design Parameters: Must have optimal length (18–25 bp), balanced GC content (40-60%), and correctly calculated melting temperatures. ***(1.5 marks)***

Must avoid secondary structures (hairpins) and primer-dimer formations. ***(1 mark)***

34. Discuss the scope and significance of Bioinformatics in modern genomic research. **[U] / [CO4]**

Ans: Scope: Merges biology, computer science, and statistics for database management, sequence analysis, and structural modelling. ***(1.5 marks)***

Big Data Significance: Essential for storing, processing, and interpreting the massive terabytes of raw data generated by NGS. ***(1 mark)***

Applications: Powers variant prediction tools for diagnostics, and aids in drug discovery and personalised medicine. ***(1.5 marks)***

35. Detail the utility of literature databases like PubMed in genomics [U] / [CO4]

Ans: Repository: Acts as the central, searchable repository for peer-reviewed biomedical literature. ***(1 mark)***

Validation: Allows researchers to perform literature mining to check if newly discovered variants have been clinically reported elsewhere. ***(1.5 marks)***

Integration: Links raw sequence data (from databases like GenBank) directly to the published biological research explaining its function. ***(1.5 marks)***

36. What is meant by "Genome Annotation"? Distinguish between structural and functional annotation. [U] / [CO4]

Ans: Definition: The process of identifying the locations of genes in a genome and assigning biological meaning to them. ***(1 mark)***

Structural Annotation: Locating the physical boundaries of genes—identifying precise locations of exons, introns, and regulatory promoters. ***(1.5 marks)***

Functional Annotation: Assigning biological roles to those genes, identifying protein domains, and mapping them to metabolic pathways. ***(1.5 marks)***

[4x8 = 32]